Statistics & Probability

Section 1: Summary Statistics & Outliers

1 The mean of five numbers is 24. When a sixth number is added, the mean becomes 26. What is the sixth number?

a. 32

b. 34

c. 36

d. 38

2 What is the median of the data set: $\{14, 8, 22, 19, 11, 25, 17, 30\}$?

a. 17

b. 18

c. 19

d. 20.5

3 An outlier is defined as any data point greater than $Q_3 + 1.5 \times \text{IQR}$. If $Q_1 = 20$ and $Q_3 = 35$, which value is considered an outlier?

a. 50

b. 55

c. 58

d. 60

Section 2: Normal Distributions & Standard Deviation

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4 Exam scores are normally distributed with $\mu = 75$ and $\sigma = 6$. What percentage of scores fall between 69 and 81?

a. 50%

b. 68%

c. 95%

d. 99.7%

5 If a test score of 88 has a z-score of 2.5 with standard deviation $\sigma = 4$, what is the mean score μ?

a. 76

b. 78

c. 80

d. 82

6 Which change will increase the standard deviation of a dataset?

a. Adding 10 to all data points

b. Multiplying all data points by 2

c. Subtracting the mean from each data point

d. Dividing all data points by 3

Section 3: Compound & Conditional Probability

7 A fair six-sided die is rolled twice. What is the probability that both rolls result in a prime number?

a. $\frac{1}{4}$

b. $\frac{1}{3}$

c. $\frac{1}{2}$

d. $\frac{1}{6}$

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8 If $P(A) = 0.5$, $P(B) = 0.6$, and $P(A \cup B) = 0.8$, what is $P(A \cap B)$?

a. 0.1

b. 0.2

c. 0.3

d. 0.4

9 A box contains 5 red balls and 7 black balls. Two balls are drawn in succession without replacement. What is the probability that both balls are red?

a. $\frac{5}{33}$

b. $\frac{5}{36}$

c. $\frac{25}{144}$

d. $\frac{1}{6}$

Section 4: Permutations & Combinations

10 How many distinct 3-person executive committees (President, Vice-President, Treasurer) can be elected from a club of 10 members?

a. 120

b. 240

c. 720

d. 1000

11 How many subsets of 3 elements can be formed from a set of 8 elements?

a. 24

b. 56

c. 112

d. 336

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12 In how many ways can 4 math books and 3 science books be arranged on a shelf if all math books must be grouped together and all science books must be grouped together?

a. 144

b. 288

c. 576

d. 5040

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